Semester VI **B. Tech CSE (AI)**

BCSAI 604: NATURAL LANGUAGE PROCESSING

3L + 0T + 0P + 3C **MM 100**

Unit 1: Overview And Morphology

Introduction, Models -and Algorithms , Regular Expressions Basic Regular Expression Patterns, Finite State Automata Morphology, Inflectional Morphology, Derivational Morphology, Finite-State Morphological Parsing , Porter Stemmer

Unit 2: Word Level And Syntactic Analysis

N-grams Models of Syntax, Counting Words, Unsmoothed N-grams, Smoothing, Back off Deleted Interpolation, Entropy , English Word Classes , Tag sets for English, Part of Speech Tagging, Rule Based Part of Speech Tagging, Stochastic Part of Speech Tagging, Transformation-Based Tagging

Unit 3: Context Free Grammars

Context Free Grammars for English Syntax, Context- Free Rules and Trees, Sentence- Level Constructions, Agreement, Sub Categorization, Parsing, Top-down, Early Parsing, feature Structures Probabilistic Context, Free Grammars

Unit 4: Semantic Analysis

Representing Meaning, Meaning Structure of Language, First Order Predicate Calculus, Representing Linguistically Relevant Concepts –Syntax, Driven Semantic Analysis, Semantic Attachments, Syntax, Driven Analyzer, Robust Analysis, Lexemes and Their Senses, Internal Structure, Word Sense Disambiguation, Information Retrieval

Unit 5: Language Generation And Discourse Analysis

Discourse -Reference Resolution , Text Coherence, Discourse Structure, Coherence, Dialog and Conversational Agents , Dialog Acts, Interpretation, Conversational Agents, Machine Translation Transfer Metaphor, Interlingua, Statistical Approaches

Text /Reference Books:

1. Daniel Jurafsky and James H Martin,"Speech and Language Processing: An introduction to Natural Language Processing, Computational Linguistics and Speech Recognition", Prentice Hall, 2nd Edition, 2008.

2. C. Manning and H. Schutze, "Foundations of Statistical Natural Language Processing", MIT Press. Cambridge, MA:,1999

3. James Allen, Bejamin/cummings, "Natural Language Understanding", 2nd edition, 1995